

Drew Council

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Experience

Senior Software Engineer

Nov 2024 — Present

Dirac, Inc.

New York, NY

- **Led implementation of a Rust-based in-browser 3D CAD viewer** that transforms complex spatial backend data into responsive, interactive visualizations for manufacturing operations.
- **Built GPU-accelerated assembly sequencing and path-planning algorithms** with custom OpenCL kernels, accelerating core automation workflows by over 10,000x and saving over \$100,000 annually.
- **Owned application-wide database migrations** across production Go microservices, designing validation and rollback paths that reduced post-migration data error rates by 80%.
- **Established an LGTM observability stack** with structured logging, metrics, and alerting across Go, Python, and C++ services, cutting production incident response times in half.
- **Engineered scalable backend services** to support the platform's core automated work instructions engine.

Software Engineer

Jan 2022 — May 2024

BotBuilt Robotics

Durham, NC

- **Worked with others on an agile team** to develop software for robotic construction of houses.
- **Headed company DevOps**, including managing CI/CD, developer tools, and embedded software deployment.
- **Designed embedded hardware** for a ROS2 network, providing services for actuator and sensor control.
- **Integrated AWS hosting** for automated Docker image builds with custom logging and error reporting.

Software Subteam Lead

Jun 2020 — Jun 2022

Duke University Robotics Club

Durham, NC

- **Competed in annual RoboSub robot competition**, where we designed a fully autonomous submarine robot.
- **Coordinated a 25+ member agile environment team** using ROS, Docker, and Git to manage a shared codebase.
- **Implemented PID, Sensor Fusion, Computer Vision, and SMACH** to improve robot accuracy and capability.
- Earned 1st in Propulsion System, 3rd in Sensor optimization in 2021; 1st in technical report in 2021 and 2022.

Projects

Duke General Robotics Lab, Cell Robots Research

Feb 2023 — Present

- Designed omnidirectional grounded robot swarm for research in distributed control and communication.
- Researched algorithms to control a swarm of robots to perform tasks with low communication and orchestration.

Duke Product Design, "Aelevate" Bike Trainer

Sep 2022 — Dec 2022

- Used PlatformIO to write C++ embedded software to read sensors and control motors on a tilting bike trainer.
- Created a serial interface to communicate with a custom GUI application for touchscreen user control.

Duke Computer Engineering, FPGA Typeracer-style Arcade Game

Sep 2022 — Dec 2022

- Used Verilog to implement a MIPS-like pipelined processor to process serial input and generate VGA graphics.
- Implemented a head-to-head typing arcade game including display and keyboard drivers on an FPGA.

Duke Engineering First Year Design, Room Availability Detection System

Sep 2020 — May 2021

- Used a Raspberry Pi and Python to analyze and error correct PIR sensor data and detect a person's presence.
- Communicated this sensor data over a custom web API to publish room status and use statistics to a website.

Education

Duke University

Durham, NC

Bachelor of Computer Science, Electrical & Computer Engineering

Sep 2020 — May 2024

- Completed classes in Data Structures/Algorithms, Computer Architecture, Networking, and Robotics.

Skills

Go, Python, Rust, C, C++, TypeScript, OpenCL, SQL, DuckDB, Linux, NixOS, Docker, Git, Bash, Arduino, Verilog, Nushell, AWS